MA385 Part 1: Solving nonlinear equations

1.6: Fixed Point Iteration

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28 September, 2026

Newton's method can be considered to be a special case of a very general approach called *Fixed Point Iteration* (**FPI**) or *Simple Iteration*.

The basic idea is:

If we want to solve $f(x) = 0$ in $[a, b]$, find a function $g(x)$ such that, if τ is such that $f(\tau) = 0$, then $g(\tau) = \tau$. Choose x_0 and set $x_{k+1} = g(x_k)$ for $k = 0, 1, 2, \ldots$.

0. Outline

1 Introduction

2 How not to choose g

3 Fixed points and contractions

- Fixed Point

4 How many iterations?

5 Newton's method as a FPI

6 Exercises

For more details, see Section 1.4 (Relaxation and Newton's method) of Süli and Mayers, *An Introduction to Numerical Analysis*

Also, Chapter 3 of Epperson:
https://search.library.nuigalway.ie/permalink/f/3b1kce/TN_cdi_askewsholts_vlebooks_9781118730966

1. Introduction

Yet again, we want to solve

Given a function $f(x)$, find $\tau \in [a, b]$ such that $f(\tau) = 0$.

Again, we'll try to find a sequence $\{x_0, x_1, \ldots, x_k, \ldots\}$, such that $x_k \to \tau$ as $k \to \infty$.

In this section, we'll consider *one-point* methods, which, like Newton's method, compute x_{k+1} just from x_k.[1]

The Method is called **Fixed Point Iteration** (FPI):

- Choose a function g such that, if $f(\tau) = 0$, then τ is a fixed point of g.
- Choose x_0, and then iterate with $x_{k+1} = g(x_k)$.

[1] In the version presented in class, I incorrectly referred to these as *one-step* method.

Example 1.6.1

Suppose $f(x) = e^x - 2x - 1$ and we are trying to find a solution to $f(x) = 0$ in $[1, 2]$. Then we can take $g(x) = \ln(2x + 1)$.

If we take $x_0 = 1$, then we get the following sequence:

k	x_k	$\lvert f(x_k)\rvert$	$\lvert\tau - x_k\rvert$
0	1.0000	0.2817	2.5643e-01
1	1.0986	0.1972	1.5782e-01
2	1.1623	0.1273	9.4148e-02
3	1.2013	0.0781	5.5092e-02
4	1.2246	0.0464	3.1868e-02
5	1.2381	0.0271	1.8310e-02
6	1.2460	0.0157	1.0479e-02
⋮	⋮	⋮	
10	1.2553	0.0017	1.1079e-03

2. How not to choose g

We have to be quite careful with this method: **not every choice is g is suitable**.

For example, suppose we want the solution to $f(x) = x^2 - 2 = 0$ in $[1, 2]$. We could choose $g(x) = x^2 + x - 2$. Then, if take $x_0 = 1$ we get the sequence:

2. How not to choose g

Before we do that in a formal way, consider the following...

Example 1.6.2

Use the Mean Value Theorem to show that the fixed point method $x_{k+1} = g(x_k)$ converges if $|g'(x)| < 1$ for all x near the fixed point.

2. How not to choose g

This previous example is useful in two ways:

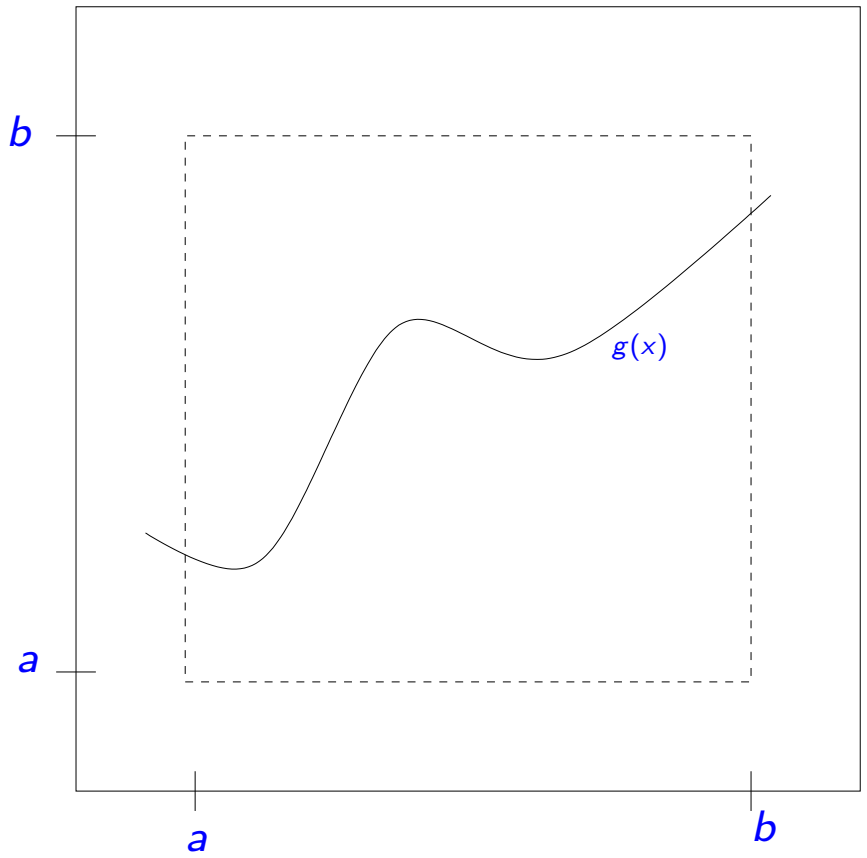
1. It introduces the tricks of using that $g(\tau) = \tau$ & $g(x_k) = x_{k+1}$.
2. It leads us towards the **contraction mapping theorem**.

Definition: fixed point

A function $g : \mathbb{R} \to \mathbb{R}$ is said to have a **fixed** point at $x = \tau$ if $g(\tau) = \tau$.

Theorem 1.6.1 (Fixed Point Theorem)

Suppose the function g is continuous on $[a, b]$, and $a \leq g(x) \leq b$ for all $x \in [a, b]$. Then $g(x)$ has a *fixed point* in $[a, b]$.



Definition: Contraction

Let g be a continuous function defined on $[a, b]$. If there is a number $L \in (0, 1)$ such that

$$|g(\alpha) - g(\beta)| \leq L|\alpha - \beta| \text{ for all } \alpha, \beta \in [a, b], \qquad (1)$$

then we say that g is a **contraction** on $[a, b]$.

Theorem 1.6.2 (Contraction Mapping Theorem)

Suppose that the function g is a real-valued, defined, continuous, and

(a) maps every point in $[a, b]$ to some point in $[a, b]$, and

(b) is a contraction on $[a, b]$,

then

(i) $g(x)$ has a fixed point $\tau \in [a, b]$,

(ii) the fixed point is unique,

(iii) the sequence $\{x_k\}_{k=0}^{\infty}$ defined by $x_0 \in [a, b]$ and $x_k = g(x_{k-1})$ for $k = 1, 2, \ldots$ converges at least linearly to τ.

4. How many iterations?

The algorithm generates as sequence $\{x_0, x_1, \ldots, x_k\}$. Eventually we must stop. Suppose we want the solution to be accurate to say 10^{-6}, how many steps are needed? That is, how big do we need to take k so that

$$|x_k - \tau| \leq 10^{-6}?$$

The answer is obtained by first showing that

$$|\tau - x_k| \leq \frac{L^k}{1-L}|x_1 - x_0|. \tag{2}$$

4. How many iterations?

Example 1.6.3

Suppose we are using FPI to find the fixed point $\tau \in [1, 2]$ of $g(x) = \ln(2x+1)$ with $x_0 = 1$, and we want $|x_k - \tau| \leq 10^{-6}$, then we can use (2) to determine the number of iterations required.

5. Newton's method as a FPI

Newton's method can be thought of as an example of a fixed point method, where we take

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

However, we know that, when Newton's Method converges it does so quadratically, whereas FPI converges (at least linearly).

5. Newton's method as a FPI

Let's remind ourselves of the definition:

- We have a sequence of numbers ε_0, ε_1, ..., such that $\lim_{k\to\infty} \varepsilon_k = 0$.
- These bound the errors: $|\tau - x_k| \leq \varepsilon_k$ Let $\tau = \lim_{k\to\infty} x_k$.
- We know that $\lim_{k\to\infty} x_k = \tau$
- Then we say that the sequence $\{x_k\}_{k=0}^{\infty}$ converges *with at least order* q if

$$\lim_{k\to\infty} \frac{\varepsilon_{k+1}}{(\varepsilon_k)^q} = \mu,$$

  for some constant μ.

For $q = 1$ we get linear convergence. If $q = 2$, we say it is *quadratic*.

5. Newton's method as a FPI

Suppose that we have a convergent Fixed Point Method, $x_{k+1} = g(x_k)$, but with the additional property that $g'(\tau) = 0$. Then, in fact, FPI converges (at least) quadratically):

5. Newton's method as a FPI

Equipped with this, one can show that, in the FPI setting, Newton converges quadratically. See Exercise.

6. Exercises

Exercise 1.6.1

Is it possible for g to be a contraction on $[a, b]$ but not have a fixed point in $[a, b]$? Give an example to support your answer.

Exercise 1.6.2

Show that $g(x) = \ln(2x + 1)$ is a contraction on $[1, 2]$. Give an estimate for L. (Hint: Use the Mean Value Theorem).

Exercise 1.6.3

Suppose we wish to numerically estimate the famous *golden ratio*, $\tau = (1+\sqrt{5})/2$, which is the positive solution to $x^2 - x - 1$. We could attempt to do this by applying fixed point iteration to the functions $g_1(x) = x^2 - 1$ or $g_2(x) = 1 + 1/x$ on the region $[3/2, 2]$.

(i) Show that g_1 is *not* a contraction on $[3/2, 2]$.

(ii) Show that g_2 *is* a contraction on $[3/2, 2]$, and give an upper bound for L.

Exercise 1.6.4

In class we saw that if $g(\tau) = \tau$, and the fixed point method given by

$$x_{k+1} = g(x_k),$$

converges to the point τ (where $g(\tau) = \tau$), and if $g'(\tau) = 0$, then the method converges quadraticially.
Show that, in fact if

$$g'(\tau) = g''(\tau) = \cdots = g^{(p-1)}(\tau) = 0,$$

then it converges with order p.

Exercise 1.6.5 (Based on Q2(d), Semester 1 exam, 25/26)

Show that Newton's method can be written as a fixed-point method.
Use the result on Slide 17 to show it converges quadratically.